

Microinjector Calibration for Use with iPSC-derived GABAergic Interneurons

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Neurovascular Unit from human iPSC-derived neurons

- Bellan Lab has one focus on human iPSC derived GABAergic neurons that can model the neurovascular unit (NVU), which can be observed & quantified through their signaling pathways, involving Nitric Oxide (NO) and Glutamate
- The NVU works as a system between neurons & blood vessels where neurons can stimulate vasodilation of the blood vessels

Current Model Involving the NVU

- The NVU can be observed in vitro by quantifying NO in a signaling pathway, where glutamate stimulates neuronal activity, which leads to nNOS producing NO from NMDAR1 activation.

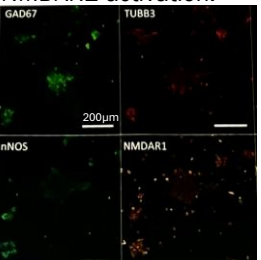


Fig. 1 nNOS & NMDAR1 iPSC-derived neurons with immunofluorescence

To observe this pathway closer a nitric oxide sensor & a glutamate "spritzer" can be integrated to stimulate the nNOS+ GABAergic neurons

Objective

Calibrate the Openspritzer (open source microinjector) to understand what pulse duration & pressure is needed for nanoliter injections

Integrate the Openspritzer into the NVU in vitro model with the GABAergic neurons along with the NO sensor

Process of Operating the Openspritzer



Fig. 3 Tip of glass capillary under 4x microscope for calibration protocol. Fragility led to changing Inner Diameter size from 2 microns to 10 microns

To view the calibration is in optimal conditions (no ballooning effect), glass capillary must be observed under a microscope & have access to compressed air for pressure.



Fig. 4 Components are as follows: (1) Air input, (2) Pressure gauge, (3) potentiometer, (4) DPDT switch, (5) foot pedal, (6) Air output, (7) LED, & (8) Pressure gauge

Protocol for the calibration included injecting fluid into mineral oil to record the radius of the sphere to determine the volume ejected. Multiple trials were done with varying pressure & pulse duration. Due to minute pulse durations in the millisecond range, parameters in Arduino were made between trials to ensure accuracy.



Fig. 6 Openspritzer in biosafety cabinet under microscope with the tip under 97 well plate

Following the calibration, discussion on how to position the micromanipulator for the Openspritzer underneath the microscope in a 96 well needed to be determined.

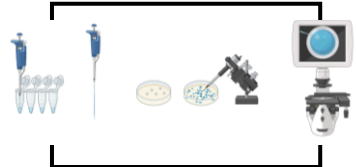


Fig. 5 Steps following calibration protocol, images from biorender.com

DIAM.	RADIUS	PICOLITER VOLUME
10 µm	5	52
20	10	14.2
30	15	33.6
40	20	65.6
50	25	113
60	30	144
70	35	180
80	40	221
90	45	269
100	50	325
110	55	389
120	60	460
130	65	538
140	70	625
150	75	721
160	80	825
170	85	938
180	90	1060
190	95	1190
200	100	1325
210	105	1465
220	110	1610
230	115	1760
240	120	1915
250	125	2075
260	130	2240
270	135	2410
280	140	2585
290	145	2765
300	150	2950
310	155	3140
320	160	3335
330	165	3535
340	170	3740
350	175	3950
360	180	4165
370	185	4385
380	190	4610
390	195	4840
400	200	5075
410	205	5315
420	210	5560
430	215	5810
440	220	6065
450	225	6325
460	230	6590
470	235	6860
480	240	7135
490	245	7415
500	250	7700

Fig. 8 Volume chart for calibration of the Openspritzer derived from Picospritzer manual (WPI)

Supplementary Work for NVU model

- 3d printing a data logger box to store component
- Thawing out IPSC's from liquid nitrogen cell storage system to an incubator for investigations



Fig. 7 IPSC thaw protocol from cell storage to centrifuge for cell pellets & place in 75 plate to incubate

Openspritzer Calibration Curve

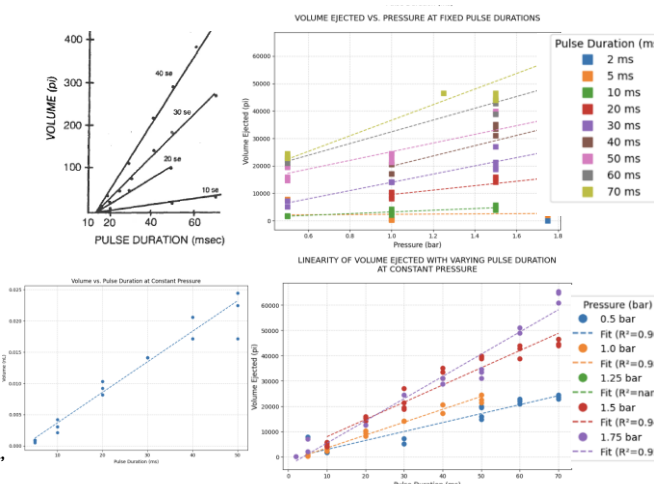


Fig. 9 Calibration curves listed from left to right: desired calibration curve, final curve with constant pressures, preliminary curve with few samples, and final curve with constant pulse duration

Preliminary Conclusion & Future Work

Using the EVOS M5000, imaging on the spheres can be seen recording volumes varying from 260pL to 13000pL. Volumes below & above this range are reproducible, however, inconsistent & have variability.

Following this work will involve injecting Glutamate into cells and incorporating both the Nitric Oxide sensor and the Openspritzer

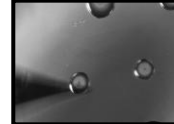


Fig. 6 The Openspritzer during calibration protocol under 10x objective

References

- Hosford, et al. *Neurosci. Biobehav. Rev.* (2019)
- Iadecola, C. (2019)
- Biorender.com for models

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